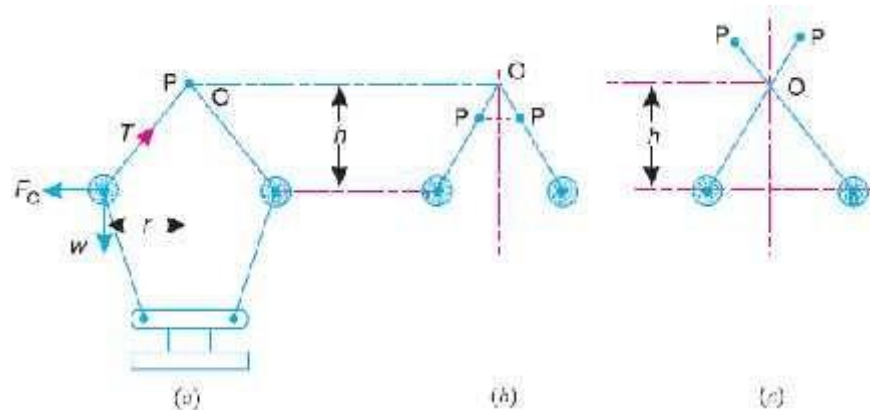


UNIT – IV

GOVERNORS

Watt Governor:



Let

m = Mass of the ball in kg.

w = Weight of the ball in newtons = $m.g$.

T = Tension in the arm in newtons.

ω = Angular velocity of the arm and ball about the spindle axis in rad/s,

r = Radius of the path of rotation of the ball *i.e.* horizontal distance from the centre of the ball to the spindle axis in metres.

F_C = Centrifugal force acting on the ball in newtons = $m.\omega^2.r$, and

h = Height of the governor in metres.

Taking moments about point O , we have

$$F_C \times h = w \times r = m.g.r$$

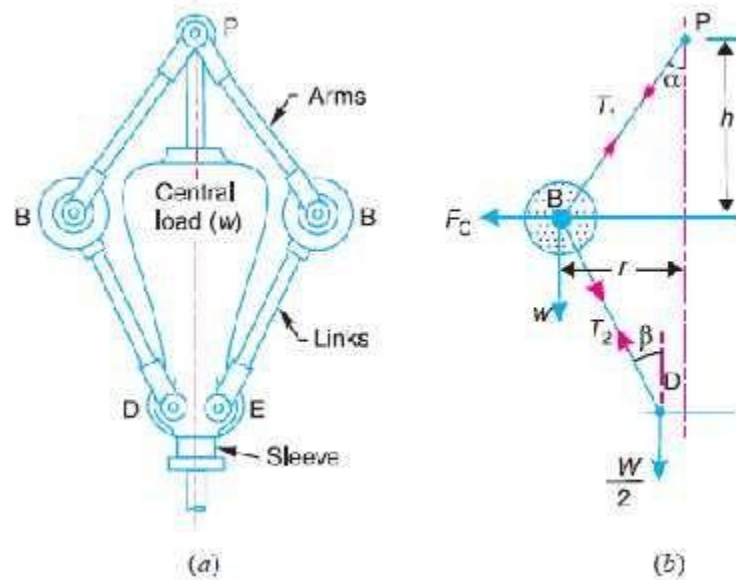
or $m.\omega^2.r.h = m.g.r$ or $h = g/\omega^2$... (i)

When g is expressed in m/s^2 and ω in rad/s, then h is in metres. If N is the speed in r.p.m., then

$$\omega = 2\pi N/60$$

$$\therefore h = \frac{9.81}{(2\pi N/60)^2} = \frac{895}{N^2} \text{ metres} \quad \dots (\because g = 9.81 \text{ m/s}^2) \dots (ii)$$

Porter Governor:



Let

m = Mass of each ball in kg.

w = Weight of each ball in newtons = $m.g$.

M = Mass of the central load in kg.

W = Weight of the central load in newtons = $M.g$.

r = Radius of rotation in metres.

h = Height of governor in metres.

N = Speed of the balls in r.p.m.,

ω = Angular speed of the balls in rad/s
 $= 2\pi N/60$ rad/s,

F_C = Centrifugal force acting on the ball
 in newtons = $m \cdot \omega^2 r$.

T_1 = Force in the arm in newtons.

T_2 = Force in the link in newtons.

α = Angle of inclination of the arm (or
 upper link) to the vertical, and

β = Angle of inclination of the link
 (or lower link) to the vertical.

$$N^2 = \frac{m + \frac{M}{2}(1+q)}{m} \times \frac{g}{h} \left(\frac{60}{2\pi} \right)^2 = \frac{m + \frac{M}{2}(1+q)}{m} \times \frac{895}{h}$$

Notes : 1. When the length of arms are equal to the length of links and the points P and D lie on the same vertical line, then

$$\tan \alpha = \tan \beta \quad \text{or} \quad q = \tan \alpha / \tan \beta = 1$$

Therefore, the equation (v) becomes

$$N^2 = \frac{(m + M)}{m} \times \frac{895}{h} \quad \dots (vi)$$

2. When the loaded sleeve moves up and down the spindle, the frictional force acts on it in a direction opposite to that of the motion of sleeve.

If F = Frictional force acting on the sleeve in newtons, then the equations (v) and (vi) may be written as

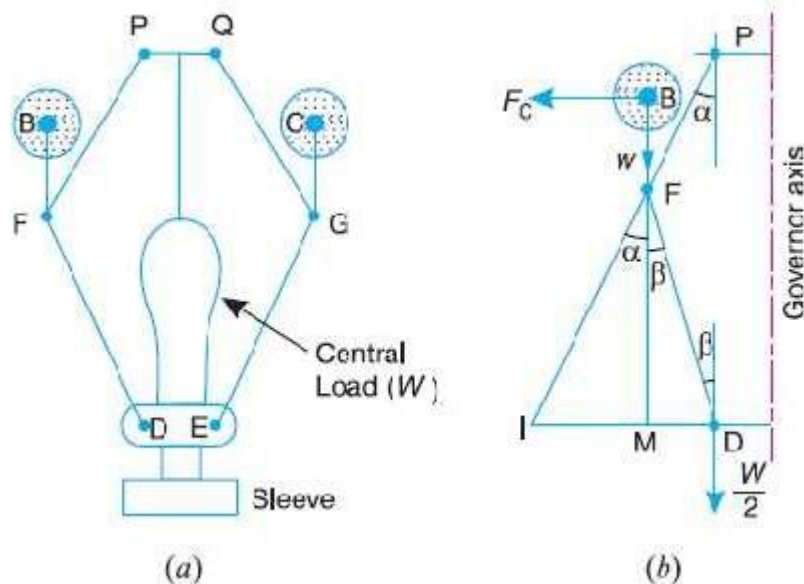
$$N^2 = \frac{m \cdot g + \left(\frac{M \cdot g \pm F}{2} \right) (1 + q)}{m \cdot g} \times \frac{895}{h} \quad \dots (vii)$$

$$= \frac{m \cdot g + (M \cdot g \pm F)}{m \cdot g} \times \frac{895}{h} \quad \dots (\text{When } q = 1) \dots (viii)$$

The + sign is used when the sleeve moves upwards or the governor speed increases and negative sign is used when the sleeve moves downwards or the governor speed decreases.

3. On comparing the equation (vi) with equation (ii) of Watt's governor (Art. 18.5), we find that the mass of the central load (M) increases the height of governor in the ratio $\frac{m + M}{m}$.

Proell Governor:



$$\omega^2 = \frac{FM}{BM} \left[\frac{m + \frac{M}{2}(1+q)}{m} \right] \frac{g}{h}$$

Substituting $\omega = 2\pi N/60$, and $g = 9.81 \text{ m/s}^2$, we get

$$N^2 = \frac{FM}{BM} \left[\frac{m + \frac{M}{2}(1+q)}{m} \right] \frac{895}{h}$$

Notes : 1. The equation (i) may be applied to any given configuration of the governor.

2. Comparing equation (iii) with the equation (v) of the Porter governor (Art. 18.6), we see that the equilibrium speed reduces for the given values of m , M and h . Hence in order to have the same equilibrium speed for the given values of m , M and h , balls of smaller masses are used in the Proell governor than in the Porter governor.

3. When $\alpha = \beta$, then $q = 1$. Therefore equation (iii) may be written as

$$N^2 = \frac{FM}{BM} \left(\frac{m+M}{m} \right) \frac{895}{h} \quad (h \text{ being in metres}) \dots (iv)$$

PROBLEMS:

Example 18.1. Calculate the vertical height of a Watt governor when it rotates at 60 r.p.m. Also find the change in vertical height when its speed increases to 61 r.p.m.

Solution. Given : $N_1 = 60 \text{ r.p.m.}$; $N_2 = 61 \text{ r.p.m.}$

Initial height

We know that initial height,

$$h_1 = \frac{895}{(N_1)^2} = \frac{895}{(60)^2} = 0.248 \text{ m}$$

Change in vertical height

We know that final height,

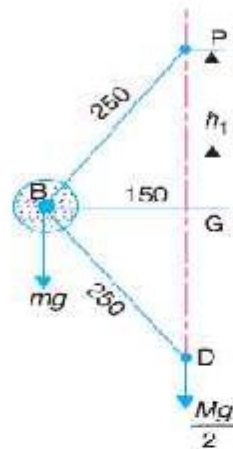
$$h_2 = \frac{895}{(N_2)^2} = \frac{895}{(61)^2} = 0.24 \text{ m}$$

$\therefore$ Change in vertical height

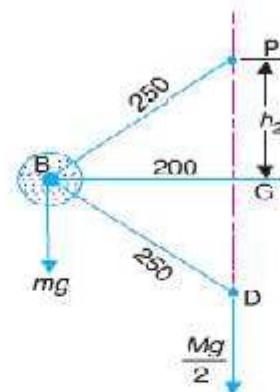
$$= h_1 - h_2 = 0.248 - 0.24 = 0.008 \text{ m} = 8 \text{ mm Ans.}$$

Example 18.2. A Porter governor has equal arms each 250 mm long and pivoted on the axis of rotation. Each ball has a mass of 5 kg and the mass of the central load on the sleeve is 25 kg. The radius of rotation of the ball is 150 mm when the governor begins to lift and 200 mm when the governor is at maximum speed. Find the minimum and maximum speeds and range of speed of the governor.

Solution. Given : $BP = BD = 250 \text{ mm} = 0.25 \text{ m}$; $m = 5 \text{ kg}$; $M = 25 \text{ kg}$; $r_1 = 150 \text{ mm} = 0.15 \text{ m}$; $r_2 = 200 \text{ mm} = 0.2 \text{ m}$



(a) Minimum position.



(b) Maximum position.

The minimum and maximum positions of the governor are shown in Fig. 18.5 (a) and (b) respectively.

Minimum speed when $r_1 = BG = 0.15 \text{ m}$

Let N_1 – Minimum speed.

From Fig. 18.5 (a), we find that height of the governor,

$$h_1 = PG = \sqrt{(PB)^2 - (BG)^2} = \sqrt{(0.25)^2 - (0.15)^2} = 0.2 \text{ m}$$

We know that

$$(N_1)^2 = \frac{m+M}{m} \times \frac{895}{h_1} = \frac{5+25}{5} \times \frac{895}{0.2} = 17900$$

$\therefore N_1 = 133.8 \text{ r.p.m. Ans.}$

Maximum speed when $r_2 = BG = 0.2 \text{ m}$

Let N_2 – Maximum speed.

From Fig. 18.5 (b), we find that height of the governor,

$$h_2 = PG = \sqrt{(PB)^2 - (BG)^2} = \sqrt{(0.25)^2 - (0.2)^2} = 0.15 \text{ m}$$

We know that

$$(N_2)^2 = \frac{m+M}{m} \times \frac{895}{h_2} = \frac{5+25}{5} \times \frac{895}{0.15} = 23867$$

$\therefore N_2 = 154.5 \text{ r.p.m. Ans.}$

Range of speed

We know that range of speed

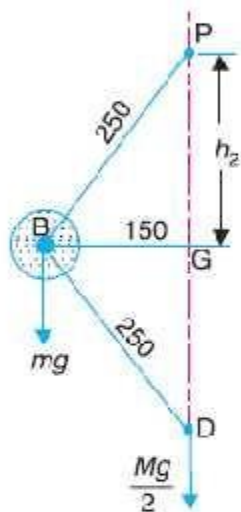
$$= N_2 - N_1 = 154.4 - 133.8 = 20.7 \text{ r.p.m. Ans.}$$

Example 18.3. The arms of a Porter governor are each 250 mm long and pivoted on the governor axis. The mass of each ball is 5 kg and the mass of the central sleeve is 30 kg. The radius of rotation of the balls is 150 mm when the sleeve begins to rise and reaches a value of 200 mm for maximum speed. Determine the speed range of the governor. If the friction at the sleeve is equivalent of 20 N of load at the sleeve, determine how the speed range is modified.

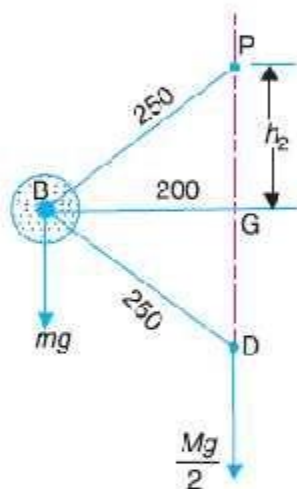
Solution. Given : $BP = BD = 250 \text{ mm}$; $m = 5 \text{ kg}$; $M = 30 \text{ kg}$; $r_1 = 150 \text{ mm}$; $r_2 = 200 \text{ mm}$

First of all, let us find the minimum and maximum speed of the governor. The minimum and maximum position of the governor is shown in Fig. 18.6 (a) and (b) respectively.

Let N_1 = Minimum speed when $r_1 = BG = 150 \text{ mm}$, and
 N_2 = Maximum speed when $r_2 = BG = 200 \text{ mm}$.



(a) Minimum position.



(b) Maximum position.

Fig. 18.6

Speed range of the governor

From Fig. 18.6 (a), we find that height of the governor,

$$h_1 = PG = \sqrt{(PB)^2 - (BG)^2} = \sqrt{(250)^2 - (150)^2} = 200 \text{ mm} = 0.2 \text{ m}$$

We know that

$$(N_1)^2 = \frac{m + M}{m} \times \frac{895}{h_1} = \frac{5 + 30}{5} \times \frac{895}{0.2} = 31\,325$$

$$\therefore N_1 = 177 \text{ r.p.m.}$$

From Fig. 18.6 (b), we find that height of the governor,

$$h_2 = PG = \sqrt{(PB)^2 - (BG)^2} = \sqrt{(250)^2 - (200)^2} = 150 \text{ mm} = 0.15 \text{ m}$$

We know that

$$(N_2)^2 = \frac{m + M}{m} \times \frac{895}{h_2} = \frac{5 + 30}{5} \times \frac{895}{0.15} = 41\,767$$

We know that speed range of the governor

$$= N_2 - N_1 = 204.4 - 177 = 27.4 \text{ r.p.m. Ans.}$$

Speed range when friction at the sleeve is equivalent of 20 N of load (i.e. when $F = 20 \text{ N}$)

We know that when the sleeve moves downwards, the frictional force (F) acts upwards and the minimum speed is given by

$$(N_1)^2 = \frac{m \cdot g + (M \cdot g - F)}{m \cdot g} \times \frac{895}{h_1}$$
$$= \frac{5 \times 9.81 + (30 \times 9.81 - 20)}{5 \times 9.81} \times \frac{895}{0.2} = 29500$$
$$\therefore N_1 = 172 \text{ r.p.m.}$$

We also know that when the sleeve moves upwards, the frictional force (F) acts downwards and the maximum speed is given by

$$(N_2)^2 = \frac{m \cdot g + (M \cdot g + F)}{m \cdot g} \times \frac{895}{h_2}$$
$$= \frac{5 \times 9.81 + (30 \times 9.81 + 20)}{5 \times 9.81} \times \frac{895}{0.15} = 44200$$
$$\therefore N_2 = 210 \text{ r.p.m.}$$

We know that speed range of the governor

$$= N_2 - N_1 = 210 - 172 = 38 \text{ r.p.m. Ans.}$$

Example 18.4. In an engine governor of the Porter type, the upper and lower arms are 200 mm and 250 mm respectively and pivoted on the axis of rotation. The mass of the central load is 15 kg, the mass of each ball is 2 kg and friction of the sleeve together with the resistance of the operating gear is equal to a load of 25 N at the sleeve. If the limiting inclinations of the upper arms to the vertical are 30° and 40° , find, taking friction into account, range of speed of the governor.

Solution. Given : $BP = 200 \text{ mm} = 0.2 \text{ m}$; $BD = 250 \text{ mm} = 0.25 \text{ m}$; $M = 15 \text{ kg}$; $m = 2 \text{ kg}$; $F = 25 \text{ N}$; $\alpha_1 = 30^\circ$; $\alpha_2 = 40^\circ$

First of all, let us find the minimum and maximum speed of the governor.

The minimum and maximum position of the governor is shown Fig. 18.7 (a) and (b) respectively.

Let N_1 = Minimum speed, and
 N_2 = Maximum speed.

From Fig. 18.7 (a), we find that minimum radius of rotation,

$$r_1 = BG = BP \sin 30^\circ = 0.2 \times 0.5 = 0.1 \text{ m}$$

Height of the governor,

$$h_1 = PG = BP \cos 30^\circ = 0.2 \times 0.866 = 0.1732 \text{ m}$$

and

$$DG = \sqrt{(BD)^2 - (BG)^2} = \sqrt{(0.25)^2 - (0.1)^2} = 0.23 \text{ m}$$

$\therefore$

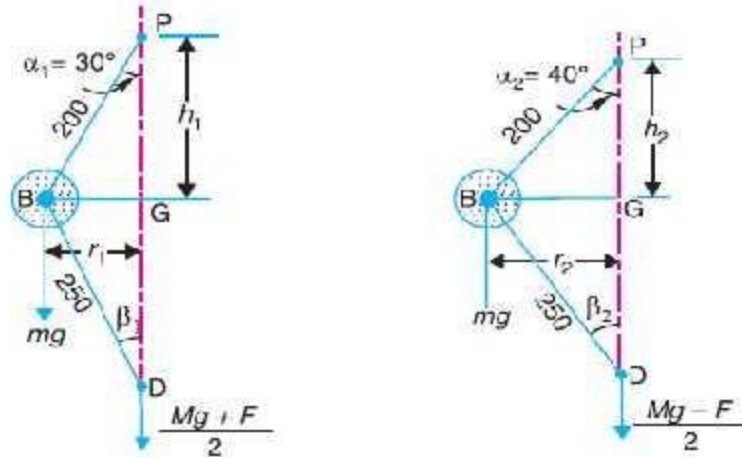
$$\tan \beta_1 = BG/DG = 0.1/0.23 = 0.4348$$

and

$$\tan \alpha_1 = \tan 30^\circ = 0.5774$$

$\therefore$

$$q_1 = \frac{\tan \beta_1}{\tan \alpha_1} = \frac{0.4348}{0.5774} = 0.753$$



All dimensions in mm.

(a) Minimum position.

(b) Maximum position.

Fig. 18.7

We know that when the sleeve moves downwards, the frictional force (F) acts upwards and the minimum speed is given by

$$\begin{aligned} (N_1)^2 &= \frac{m \cdot g + \left(\frac{M \cdot g - F}{2} \right) (1 + q_1)}{m \cdot g} \times \frac{895}{h_1} \\ &= \frac{2 \times 9.81 + \left(\frac{15 \times 9.81 - 24}{2} \right) (1 + 0.753)}{2 \times 9.81} \times \frac{895}{0.1732} = 33.596 \end{aligned}$$

$$\therefore N_1 = 183.3 \text{ r.p.m.}$$

Now from Fig. 18.7 (b), we find that maximum radius of rotation,

$$r_2 = BG = BP \sin 40^\circ = 0.2 \times 0.643 = 0.1268 \text{ m}$$

Height of the governor,

$$h_2 = PG = BP \cos 40^\circ = 0.2 \times 0.766 = 0.1532 \text{ m}$$

and

$$DG = \sqrt{(BD)^2 - (BG)^2} = \sqrt{(0.25)^2 - (0.1268)^2} = 0.2154 \text{ m}$$

$\therefore$

$$\tan \beta_2 = BG/DG = 0.1268 / 0.2154 = 0.59$$

and

$$\tan \alpha_2 = \tan 40^\circ = 0.839$$

$\therefore$

$$q_2 = \frac{\tan \beta_2}{\tan \alpha_2} = \frac{0.59}{0.839} = 0.703$$

We know that when the sleeve moves upwards, the frictional force (F) acts downwards and the maximum speed is given by

$$(N_2)^2 = \frac{m \cdot g + \left(\frac{m \cdot g + F}{2} \right) (1 + q_2)}{m \cdot g} \times \frac{895}{h_2}$$

$$= \frac{2 \times 9.81 + \left(\frac{15 \times 9.81 + 24}{2} \right) (1 + 0.703)}{2 \times 9.81} \times \frac{895}{0.1532} = 49\,236$$

$$\therefore N_2 = 222 \text{ r.p.m.}$$

We know that range of speed

$$= N_2 - N_1 = 222 - 183.3 = 38.7 \text{ r.p.m. Ans.}$$

Example 18.5. A Porter governor has all four arms 250 mm long. The upper arms are attached on the axis of rotation and the lower arms are attached to the sleeve at a distance of 30 mm from the axis. The mass of each ball is 5 kg and the sleeve has a mass of 50 kg. The extreme radii of rotation are 150 mm and 200 mm. Determine the range of speed of the governor.

Solution. Given : $BP = BD = 250 \text{ mm}$; $DH = 30 \text{ mm}$; $m = 5 \text{ kg}$; $M = 50 \text{ kg}$; $r_1 = 150 \text{ mm}$; $r_2 = 200 \text{ mm}$

First of all, let us find the minimum and maximum speed of the governor. The minimum and maximum position of the governor is shown in Fig. 18.8 (a) and (b) respectively.

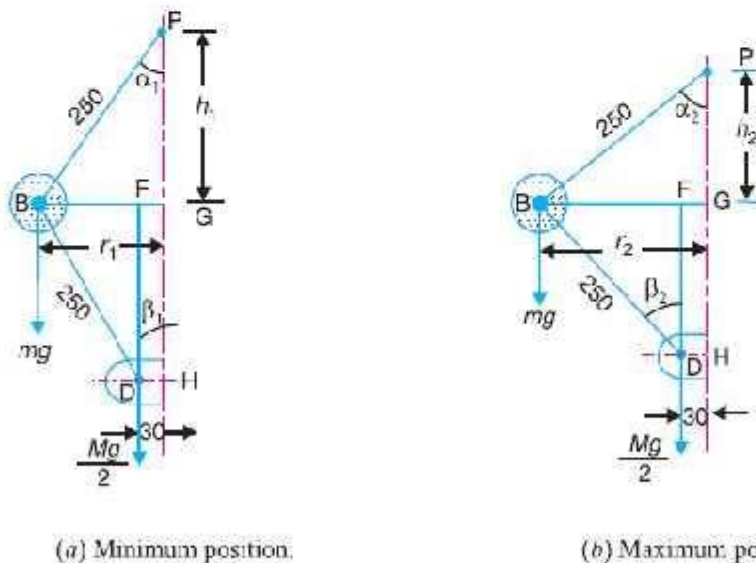


Fig. 18.8

Let N_1 = Minimum speed when $r_1 = BG = 150 \text{ mm}$; and

N_2 = Maximum speed when $r_2 = BG = 200 \text{ mm}$.

From Fig. 18.8 (a), we find that height of the governor,

$$h_1 = PG = \sqrt{(BP)^2 - (BG)^2} = \sqrt{(250)^2 - (150)^2} = 200 \text{ mm} = 0.2 \text{ m}$$

$$BF = BG - FG = 150 - 30 = 120 \text{ mm} \quad \dots (\because FG = DH)$$

and $DF = \sqrt{(DB)^2 - (BF)^2} = \sqrt{(250)^2 - (120)^2} = 219 \text{ mm}$

$\therefore \tan \alpha_1 = BG/PG = 150/200 = 0.75$

and $\tan \beta_1 = BF/DF = 120/219 = 0.548$

$\therefore q_1 = \frac{\tan \beta_1}{\tan \alpha_1} = \frac{0.548}{0.75} = 0.731$

We know that $(N_1)^2 = \frac{m + \frac{M}{2}(1 + q_1)}{m} \times \frac{895}{h_1} = \frac{5 + \frac{50}{2}(1 + 0.731)}{5} \times \frac{895}{0.2} = 43205$

$\therefore N_1 = 208 \text{ r.p.m.}$

From Fig. 18.8(b), we find that height of the governor,

$$h_2 = PG = \sqrt{(BP)^2 - (BG)^2} = \sqrt{(250)^2 - (200)^2} = 150 \text{ mm} = 0.15 \text{ m}$$

$$BF = BG - FG = 200 - 30 = 170 \text{ mm}$$

and $DF = \sqrt{(DB)^2 - (BF)^2} = \sqrt{(250)^2 - (170)^2} = 183 \text{ mm}$

$\therefore \tan \alpha_2 = BG/PG = 200/150 = 1.333$

and $\tan \beta_2 = BF/DF = 170/183 = 0.93$

$\therefore q_2 = \frac{\tan \beta_2}{\tan \alpha_2} = \frac{0.93}{1.333} = 0.7$

We know that

$$(N_2)^2 = \frac{m + \frac{M}{2}(1 + q_2)}{m} \times \frac{895}{h_2} = \frac{5 + \frac{50}{2}(1 + 0.7)}{5} \times \frac{895}{0.15} = 56683$$

$\therefore N_2 = 238 \text{ r.p.m.}$

We know that range of speed

$$= N_2 - N_1 = 238 - 208 = 30 \text{ r.p.m. Ans.}$$

Example 18.6. The arms of a Porter governor are 300 mm long. The upper arms are pivoted on the axis of rotation. The lower arms are attached to a sleeve at a distance of 40 mm from the axis of rotation. The mass of the load on the sleeve is 70 kg and the mass of each ball is 10 kg. Determine the equilibrium speed when the radius of rotation of the balls is 200 mm. If the friction is equivalent to a load of 20 N at the sleeve, what will be the range of speed for this position?

Solution. Given: $BP = BD = 300 \text{ mm}$; $DH = 40 \text{ mm}$; $M = 70 \text{ kg}$; $m = 10 \text{ kg}$; $r = BG = 200 \text{ mm}$

Equilibrium speed when the radius of rotation $r = BG = 200 \text{ mm}$

Let N = Equilibrium speed.

The equilibrium position of the governor is shown in Fig. 18.9. From the figure, we find that height of the governor,

$$\begin{aligned} h = PG &= \sqrt{(BP)^2 - (BG)^2} = \sqrt{(300)^2 - (200)^2} = 224 \text{ mm} \\ &= 0.224 \text{ m} \end{aligned}$$

$$\therefore BF = BG - FG = 200 - 40 = 160 \quad \dots (\because FG = DH)$$

and $DF = \sqrt{(DB)^2 - (BF)^2} = \sqrt{(300)^2 - (160)^2} = 254 \text{ mm}$

$$\therefore \tan \alpha = BG/PG = 200 / 224 = 0.893$$

and $\tan \beta = BF/DF = 160 / 254 = 0.63$

$$\therefore q = \frac{\tan \beta}{\tan \alpha} = \frac{0.63}{0.893} = 0.705$$

We know that

$$N_2 = \frac{m + \frac{M}{2} (1 + q)}{m} \times \frac{895}{h}$$

$$= \frac{10 + \frac{70}{2} (1 + 0.705)}{10} \times \frac{895}{0.224} = 27\,840$$

$$\therefore N_2 = 167 \text{ r.p.m.} \quad \text{Ans.}$$

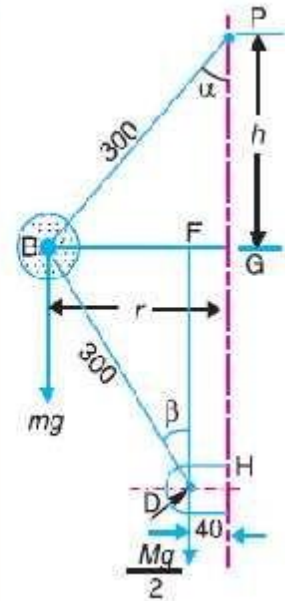
Range of speed when friction is equivalent to load of 20 N at the sleeve (i.e. when $F = 20 \text{ N}$)

Let N_1 = Minimum equilibrium speed, and

N_2 = Maximum equilibrium speed.

We know that when the sleeve moves downwards, the frictional force (F) acts upwards and the minimum equilibrium speed is given by

$$(N_1)^2 = \frac{m \cdot g + \left(\frac{M \cdot g - F}{2} \right) (1 + q)}{m \cdot g} \times \frac{895}{h}$$



$$= \frac{10 \times 9.81 + \left(\frac{70 \times 9.81 - 20}{2} \right) (1 + 0.705)}{10 \times 9.81} \times \frac{895}{0.224} = 27\,144$$

$$\therefore N_1 = 164.8 \text{ r.p.m.}$$

We also know that when the sleeve moves upwards, the frictional force (F) acts downwards and the maximum equilibrium speed is given by

$$(N_2)^2 = \frac{m \cdot g + \left(\frac{M \cdot g + F}{2} \right) (1 + q)}{m \cdot g} \times \frac{895}{h}$$

$$= \frac{10 \times 9.81 + \left(\frac{70 \times 9.81 + 20}{2} \right) (1 + 0.705)}{10 \times 9.81} \times \frac{895}{0.224} = 28\,533$$

$$\therefore N_2 = 169 \text{ r.p.m.}$$

We know that range of speed

$$= N_2 - N_1 = 169 - 164.8 = 4.2 \text{ r.p.m. Ans.}$$

Example 18.7. A loaded Porter governor has four links each 250 mm long, two revolving masses each of 3 kg and a central dead weight of mass 20 kg. All the links are attached to respective sleeves at radial distances of 40 mm from the axis of rotation. The masses revolve at a radius of 150 mm at minimum speed and at a radius of 200 mm at maximum speed. Determine the range of speed.

Solution. Given : $BP = BD = 250 \text{ mm}$; $m = 3 \text{ kg}$; $M = 20 \text{ kg}$; $PQ = DH = 40 \text{ mm}$; $r_1 = 150 \text{ mm}$; $r_2 = 200 \text{ mm}$

First of all, let us find the minimum and maximum speed of the governor.

The minimum and maximum position of the governor is shown in Fig. 18.10 (a) and (b) respectively.

Let N_1 = Minimum speed when $r_1 = BG = 150 \text{ mm}$, and

N_2 = Minimum speed when $r_2 = BG = 200 \text{ mm}$.

From Fig. 18.10 (a), we find that

$$BF = BG - FG = 150 - 40 = 110 \text{ mm}$$

$$\text{and} \quad \sin \alpha_1 = BF / BP = 110 / 250 = 0.44 \quad \text{or} \quad \alpha_1 = 26.1^\circ$$

$\therefore$ Height of the governor,

$$h_1 = OG = BG / \tan \alpha_1 = 150 / \tan 26.1^\circ = 306 \text{ mm} = 0.306 \text{ m}$$

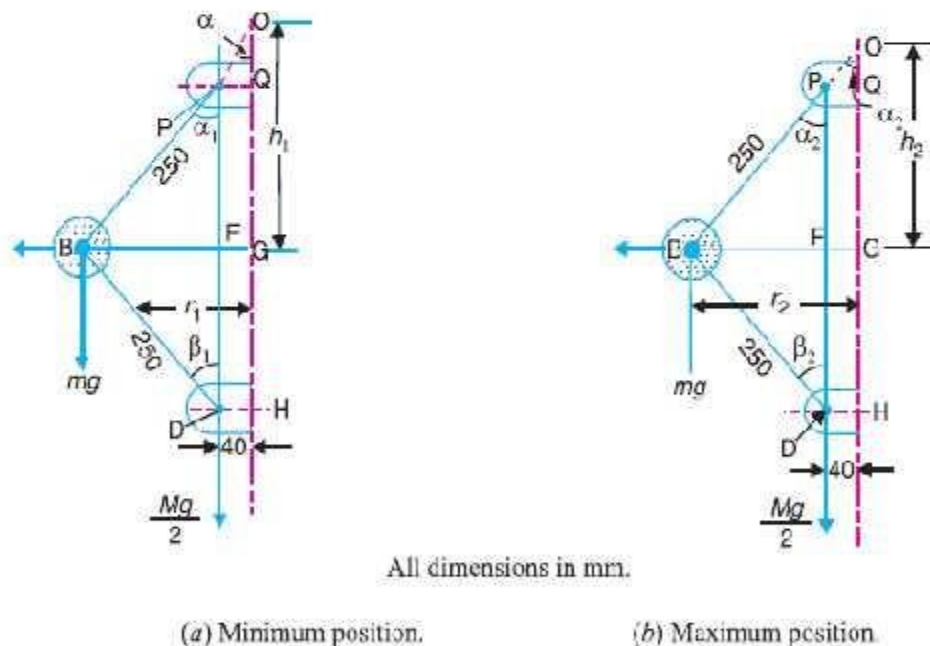


Fig. 18.10

Since all the links are attached to respective sleeves at equal distances (*i.e.* 40 mm) from the axis of rotation, therefore

$$\tan \alpha_1 = \tan \beta_1 \quad \text{or} \quad q = 1$$

We know that

$$(N_1)^2 = \frac{m + M}{m} \times \frac{895}{h_1} = \frac{3 + 20}{3} \times \frac{895}{0.306} = 22424$$

$$N_1 = 150 \text{ r.p.m.}$$

Now from Fig. 18.10 (b), we find that

$$BI' = BG - I'G = 200 - 40 = 160 \text{ mm}$$

and

$$\sin \alpha_2 = BI'/BP = 160/250 = 0.64 \quad \text{or} \quad \beta_2 = 39.8^\circ$$

$\therefore$ Height of the governor,

$$h_2 = OG = BG / \tan \alpha_2 = 200 / \tan 39.8^\circ = 240 \text{ mm} = 0.24 \text{ m}$$

In this case also,

$$\tan \alpha_2 = \tan \beta_2 \quad \text{or} \quad q = 1$$

We know that

$$(N_2)^2 = \frac{m + M}{m} \times \frac{895}{h_2} = \frac{3 + 20}{3} \times \frac{895}{0.24} = 28\,590$$

$$\therefore N_2 = 169 \text{ r.p.m.}$$

We know that range of speed

$$= N_2 - N_1 = 169 - 150 = 19 \text{ r.p.m.} \quad \text{Ans.}$$

Example 18.8. All the arms of a Porter governor are 178 mm long and are hinged at a distance of 38 mm from the axis of rotation. The mass of each ball is 1.15 kg and mass of the sleeve is 20 kg. The governor sleeve begins to rise at 280 r.p.m. when the links are at an angle of 30° to the vertical. Assuming the friction force to be constant, determine the minimum and maximum speed of rotation when the inclination of the arms to the vertical is 45° .

Solution. Given : $BP = BD = 178 \text{ mm}$; $PQ = DH = 38 \text{ mm}$;
 $m = 1.15 \text{ kg}$; $M = 20 \text{ kg}$; $N = 280 \text{ r.p.m.}$; $\alpha = \beta = 30^\circ$

First of all, let us find the friction force (F). The equilibrium position of the governor when the links are at 30° to vertical, is shown in Fig. 18.11. From the figure, we find that radius of rotation,

$$\begin{aligned} r = BG &= BI' + I'G = BP \times \sin \alpha + I'G \\ &= 178 \sin 30^\circ + 38 = 127 \text{ mm} \end{aligned}$$

and height of the governor,

$$\begin{aligned} h &= BG / \tan \alpha \\ &= 127 / \tan 30^\circ = 220 \text{ mm} = 0.22 \text{ m} \end{aligned}$$

We know that

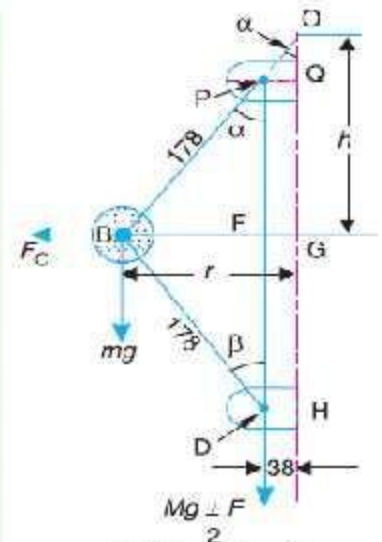
$$N^2 = \frac{m \cdot g + (Mg \pm F)}{m \cdot g} \times \frac{895}{h} \quad \dots (\because \tan \alpha = \tan \beta \text{ or } q = 1)$$

$$(280)^2 = \frac{1.15 \times 9.81 + 20 \times 9.81 \pm F}{1.15 \times 9.81} \times \frac{895}{0.22}$$

$$\begin{aligned} \text{or} \quad +F &= \frac{(280)^2 \times 1.15 \times 9.81 \times 0.22}{895} - 1.15 \times 9.81 - 20 \times 9.81 \\ &= 217.5 - 11.3 - 196.2 = 10 \text{ N} \end{aligned}$$

We know that radius of rotation when inclination of the arms to the vertical is 45° (i.e. when $\alpha = \beta = 45^\circ$),

$$\begin{aligned} r &= BG = BF + FG = BP \times \sin \alpha + FG \\ &= 178 \sin 45^\circ + 38 = 164 \text{ mm} \end{aligned}$$



All dimensions in mm.

Fig. 18.11

and height of the governor,

$$h = BG / \tan \alpha = 164 / \tan 45^\circ = 164 \text{ mm} = 0.164 \text{ m}$$

Let N_1 = Minimum speed of rotation, and

N_2 = Maximum speed of rotation.

We know that

$$\begin{aligned} (N_1)^2 &= \frac{m \cdot g + (M \cdot g - F)}{m \cdot g} \times \frac{895}{h} \\ &= \frac{1.15 \times 9.81 + (20 \times 9.81 - 10)}{1.15 \times 9.81} \times \frac{895}{0.164} = 95\,382 \end{aligned}$$

$$\therefore N_1 = 309 \text{ r.p.m. } \textbf{Ans.}$$

and

$$\begin{aligned} (N_2)^2 &= \frac{m \cdot g + (M \cdot g + F)}{m \cdot g} \times \frac{895}{h} \\ &= \frac{1.15 \times 9.81 + (20 \times 9.81 + 10)}{1.15 \times 9.81} \times \frac{895}{0.164} = 105\,040 \end{aligned}$$

$$N_2 = 324 \text{ r.p.m. } \textbf{Ans.}$$

Example 18.9. A Proell governor has equal arms of length 300 mm. The upper and lower ends of the arms are pivoted on the axis of the governor. The extension arms of the lower links are each 80 mm long and parallel to the axis when the radii of rotation of the balls are 150 mm and 200 mm. The mass of each ball is 10 kg and the mass of the central load is 100 kg. Determine the range of speed of the governor.

Solution. Given : $PF = DF = 300 \text{ mm}$; $BF = 80 \text{ mm}$; $m = 10 \text{ kg}$; $M = 100 \text{ kg}$; $r_1 = 150 \text{ mm}$; $r_2 = 200 \text{ mm}$

First of all, let us find the minimum and maximum speed of the governor. The minimum and maximum position of the governor is shown in Fig. 18.13.

Let N_1 = Minimum speed when radius of rotation, $r_1 = FG = 150 \text{ mm}$; and

N_2 = Maximum speed when radius of rotation, $r_2 = FG = 200 \text{ mm}$.

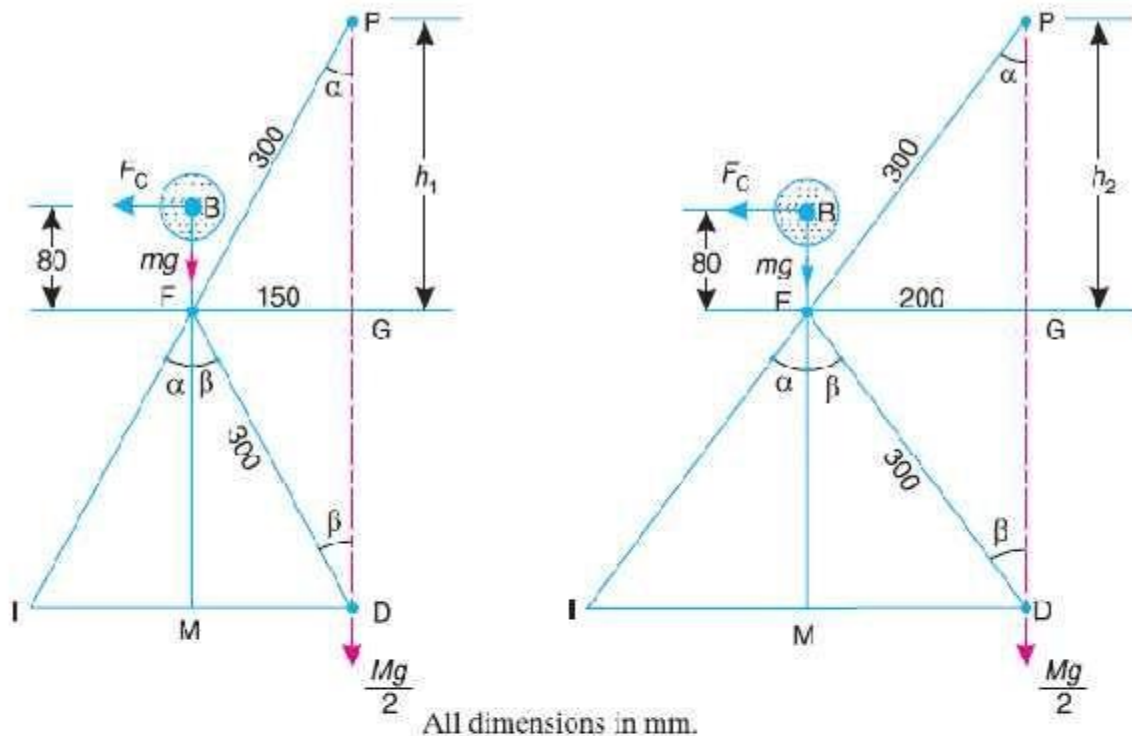
From Fig. 18.13 (a), we find that height of the governor,

$$h_1 = PG = \sqrt{(PF)^2 - (FG)^2} = \sqrt{(300)^2 - (150)^2} = 260 \text{ mm} = 0.26 \text{ m}$$

$$\text{and } FM = GD = PG = 260 \text{ mm} = 0.26 \text{ m}$$

$$\therefore BM = BF + FM = 80 + 260 = 340 \text{ mm} = 0.34 \text{ m}$$

$$\begin{aligned} \text{We know that } (N_1)^2 &= \frac{FM}{BM} \left(\frac{m + M}{m} \right) \frac{895}{h_1} \quad \dots (\because \alpha = \beta \text{ or } q = 1) \\ &= \frac{0.26}{0.34} \left(\frac{10 + 100}{10} \right) \frac{895}{0.26} = 28\,956 \quad \text{or } N_1 = 170 \text{ r.p.m.} \end{aligned}$$



(a) Minimum position.

(a) Maximum position.

Fig. 18.13

Now from Fig. 18.13 (b), we find that height of the governor,

$$h_2 = PG = \sqrt{(PF)^2 - (FG)^2} = \sqrt{(300)^2 - (200)^2} = 224 \text{ mm} = 0.224 \text{ m}$$

and

$$FM = GD = PG = 224 \text{ mm} = 0.224 \text{ m}$$

$$\therefore BM = BF + FM = 80 + 224 = 304 \text{ mm} = 0.304 \text{ m}$$

$$\text{We know that } (N_2)^2 = \frac{FM}{BM} \left(\frac{m + M}{m} \right) \frac{895}{h_2} \quad \because \alpha = \beta \text{ or } q = 1$$

$$= \frac{0.224}{0.304} \left(\frac{10 + 100}{10} \right) \frac{895}{0.224} = 32\,385 \quad \text{or} \quad N_2 = 180 \text{ r.p.m.}$$

We know that range of speed

$$= N_2 - N_1 = 180 - 170 = 10 \text{ r.p.m.} \quad \text{Ans.}$$

Example 18.10. A governor of the Proell type has each arm 250 mm long. The pivots of the upper and lower arms are 25 mm from the axis. The central load acting on the sleeve has a mass of 25 kg and the each rotating ball has a mass of 3.2 kg. When the governor sleeve is in mid position, the extension link of the lower arm is vertical and the radius of the path of rotation of the masses is 175 mm. The vertical height of the governor is 200 mm.

If the governor speed is 160 r.p.m. when in mid-position, find : 1. length of the extension link; and 2. tension in the upper arm.

Solution. Given : $PF = DF = 250 \text{ mm} = 0.25 \text{ m}$; $PQ = DH = KG = 25 \text{ mm} = 0.025 \text{ m}$
 $M = 25 \text{ kg}$; $m = 3.2 \text{ kg}$; $r = FG = 175 \text{ mm} = 0.175 \text{ m}$; $h = QG = PK = 200 \text{ mm} = 0.2 \text{ m}$; $N = 160 \text{ r.p.m}$

1. Length of the extension link

Let BF = Length of the extension link.

The Proell governor in its mid-position is shown in Fig. 18.14.

From the figure, we find that

$$FM = GH = QG = 200 \text{ mm} = 0.2 \text{ m}$$

We know that

$$N^2 = \frac{FM}{BM} \left(\frac{m + M}{m} \right) \frac{895}{h} \quad \dots (\because \alpha = \beta \text{ or } q = 1)$$

$$(160)^2 = \frac{0.2}{BM} \left(\frac{3.2 + 25}{3.2} \right) \frac{895}{0.2} = \frac{7887}{BM}$$

$$\therefore BM = 7887 / (160)^2 = 0.308 \text{ m}$$

From Fig. 18.14,

$$BF = BM - FM = 0.308 - 0.2 = 0.108 \text{ m} = 108 \text{ mm} \quad \text{Ans.}$$

2. Tension in the upper arm

Let T_1 = Tension in the upper arm.

$$PK = \sqrt{(PF)^2 - (FK)^2} = \sqrt{(PF)^2 - (FG - KG)^2}$$

$$= \sqrt{(250)^2 - (175 - 25)^2} = 200 \text{ mm}$$

$$\cos \alpha = PK/PF = 200/250 = 0.8$$

and $T_1 \cos \alpha = mg + \frac{Mg}{2} = 3.2 \times 9.81 + \frac{25 \times 9.81}{2} = 154 \text{ N}$

$$\therefore T_1 = \frac{154}{\cos \alpha} = \frac{154}{0.8} = 192.5 \text{ N} \quad \text{Ans.}$$

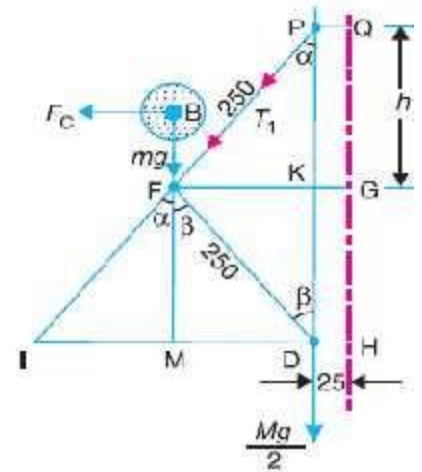


Fig. 18.14. All dimensions in mm.

Example 18.11. The following particulars refer to a Proell governor with open arms :

Length of all arms = 200 mm ; distance of pivot of arms from the axis of rotation = 40 mm ; length of extension of lower arms to which each ball is attached = 100 mm ; mass of each ball = 6 kg and mass of the central load = 150 kg. If the radius of rotation of the balls is 180 mm when the arms are inclined at an angle of 40° to the axis of rotation, find the equilibrium speed for the above configuration.

Solution. Given : $PI' = DI' = 200 \text{ mm}$; $PQ = DK = HG = 40 \text{ mm}$; $BF = 100 \text{ mm}$; $m = 6 \text{ kg}$; $M = 150 \text{ kg}$; $r = JG = 180 \text{ mm} = 0.18 \text{ m}$; $\alpha = \beta = 40^\circ$

Let N = Equilibrium speed.

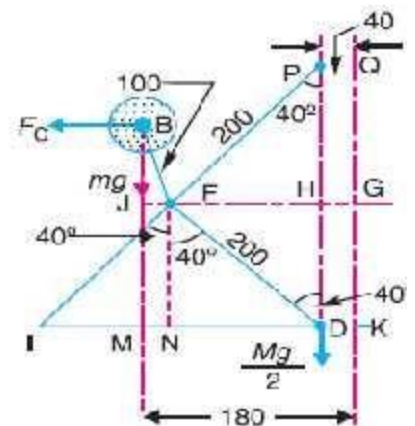


Fig. 18.15. All dimensions in mm

From the equilibrium position of the governor, as shown in Fig. 18.15, we find that

$$\begin{aligned} PH &= PF \times \cos 40^\circ \\ &= 200 \times 0.766 = 153.2 \text{ mm} \\ &= 0.1532 \text{ m} \end{aligned}$$

and $FH = PF \times \sin 40^\circ = 200 \times 0.643 = 128.6 \text{ mm}$

$\therefore JF = JG - HG - FH = 180 - 40 - 128.6 = 11.4 \text{ mm}$

and $BJ = \sqrt{(BF)^2 - (JF)^2} = \sqrt{(100)^2 - (11.4)^2} = 99.4 \text{ mm}$

We know that $BM = BJ + JM = 99.4 + 153.2 = 252.6 \text{ mm}$ $\dots (\because JM = HD = PH)$

$$IM = IN = NM = FH = JF = 128.6 + 11.4 = 140 \text{ mm}$$

$\dots (\because IN = ND = FH)$

and $ID = IN + ND = 2 \times IN = 2 \times FH = 2 \times 128.6 = 257.2 \text{ mm}$

Now taking moments about the instantaneous centre I ,

$$F_C \times BM = m \cdot g \times IM + \frac{M \cdot g}{2} \times ID$$

$$F_C \times 252.6 = 6 \times 9.81 \times 140 + \frac{150 \times 9.81}{2} \times 257.2 = 196125$$

$\therefore F_C = \frac{196125}{252.6} = 776.4 \text{ N}$

We know that centrifugal force (F_C),

$$776.4 = m \cdot \omega^2 r = 6 \left(\frac{2\pi N}{60} \right)^2 0.18 = 0.012 N^2$$

$\therefore N^2 = \frac{776.4}{0.012} = 64700 \quad \text{or} \quad N = 254 \text{ r.p.m. Ans.}$

Example 18.12. A Proell governor has all four arms of length 305 mm. The upper arms are pivoted on the axis of rotation and the lower arms are attached to a sleeve at a distance of 38 mm from the axis. The mass of each ball is 4.8 kg and are attached to the extension of the lower arms which are 102 mm long. The mass on the sleeve is 45 kg. The minimum and maximum radii of governor are 165 mm and 216 mm. Assuming that the extensions of the lower arms are parallel to the governor axis at the minimum radius, find the corresponding equilibrium speeds.

Solution. Given : $PF = DF = 305 \text{ mm}$; $DH = 38 \text{ mm}$; $BF = 102 \text{ mm}$; $m = 4.8 \text{ kg}$; $M = 54 \text{ kg}$

Equilibrium speed at the minimum radius of governor

The radius of the governor is the distance of the point of intersection of the upper and lower arms from the governor axis. When the extensions of the lower arms are parallel to the governor axis, then the radius of the governor (FG) is equal to the radius of rotation (r_1).

Fig. 18.16

$$\sin \alpha = \frac{FG}{FP} = \frac{165}{305} = 0.541$$

$\therefore \alpha = 32.75^\circ$

and $\tan \alpha = \tan 32.75^\circ = 0.6432$

$$\therefore \tan \beta = \tan 24.6^\circ = 0.4578$$
$$q = \frac{\tan \beta}{\tan \alpha} = \frac{0.4578}{0.6432} = 0.712$$
$$h = PG = \sqrt{(PF)^2 - (FG)^2} = \sqrt{(305)^2 - (165)^2} = 256.5 \text{ mm} = 0.2565 \text{ m}$$

$$MD = FK = FG - KG = 165 - 38 = 127 \text{ mm}$$

We know that

$$(N_1)^2 = \frac{FM}{BM} \left[\frac{m + \frac{M}{2} (1 + q)}{m} \right] \frac{895}{h}$$

$$= \frac{0.277}{0.379} \left[\frac{4.8 + \frac{54}{2} (1 + 0.712)}{4.8} \right] \frac{895}{0.2565} = 27109$$

Equilibrium speed at the maximum radius of governor

Let N_2 = Equilibrium speed at the maximum radius of governor, i.e., when $F_1 G_1 = r_5 = 216$ mm.

and

$$\tan \gamma = MD/BM = 127/379 = 0.335 \quad \text{or} \quad \gamma = 18.5^\circ$$

2

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$\beta_1 = 35.7^\circ$

and

$$\gamma = \beta - \gamma_1 = \beta_1 \text{ or } \gamma_1 - \gamma = \beta + \beta_1 = 18.5^\circ - 24.6^\circ + 35.7^\circ = 29.6^\circ$$

$\therefore$ Radius of rotation.

From Fig. 18.17, we find that

$$F_1 N_1 = F_1 D_1 \times \cos \beta_1 = 305 \times \cos 35.7^\circ = 248 \text{ mm} = 0.248 \text{ m}$$

$$N_1 D_1 = F_1 D_1 \times \sin \beta_1 = 305 \times \sin 35.7 = 178 \text{ mm} = 0.178 \text{ m}$$

$$I_1 D_1 = I_1 N_1 + N_1 D_1 = 0.249 + 0.178 = 0.427 \text{ m}$$

$$M_1 D_1 = B_1 D_1 \sin \gamma_1 = 400 \sin 29.6^\circ = 198 \text{ mm} = 0.198 \text{ m}$$

$$I_1 M_1 - I_1 D_1 = M_1 D_1 = 0.427 \quad 0.198 - 0.229 \text{ m}$$

We know that centrifugal force,

$$F_C = m(\omega_2)^2 r_2 = 4.8 \left(\frac{2\pi N_2}{60} \right)^2 [0.2356 - 0.0124 (N_2)^2]$$

Now taking moments about point I_1 ,

$$F_C \times B_1 M_1 - m \cdot g \times I_1 M_1 + \frac{M \cdot g}{2} \times I_1 D_1$$

$$0.0124 (N_2)^2 \times 0.348 - 4.8 \times 9.81 \times 0.229 + \frac{54 \times 9.81}{2} \times 0.427$$

$$0.0043 (N_s)^2 = 10.873 + 113.1 = 123.883$$

$$\therefore (N_2)^2 = \frac{123.883}{0.0043} = 28\,810 \quad \text{or} \quad N_2 = 170 \text{ r.p.m.} \quad \text{Ans.}$$

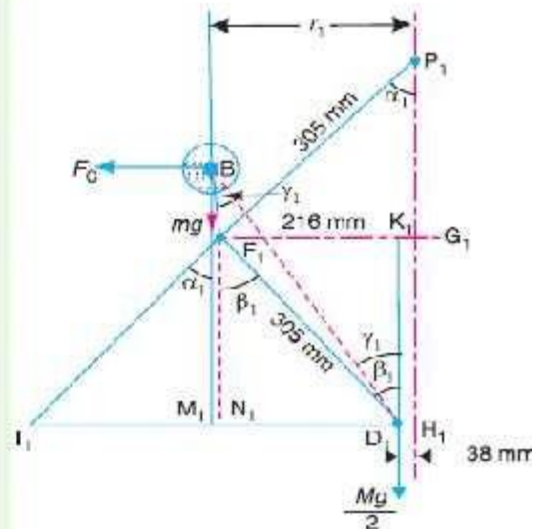


Fig. 18.17

Example 18.13. A Hartnell governor having a central sleeve spring and two right-angled bell crank levers moves between 290 r.p.m. and 310 r.p.m. for a sleeve lift of 15 mm. The sleeve arms and the ball arms are 80 mm and 120 mm respectively. The levers are pivoted at 120 mm from the governor axis and mass of each ball is 2.5 kg. The ball arms are parallel to the governor axis at the lowest equilibrium speed. Determine : 1. loads on the spring at the lowest and the highest equilibrium speeds, and 2. stiffness of the spring.

Solution. Given : $N_1 = 290$ r.p.m. or $\omega_1 = 2\pi \times 290/60 = 30.4$ rad/s ; $N_2 = 310$ r.p.m. or $\omega_2 = 2\pi \times 310/60 = 32.5$ rad/s ; $h = 15$ mm = 0.015 m ; $y = 80$ mm = 0.08 m ; $x = 120$ mm = 0.12 m ; $r = 120$ mm = 0.12 m ; $m = 2.5$ kg

1. **Loads on the spring at the lowest and highest equilibrium speeds**

Let S_1 = Spring load at lowest equilibrium speed, and
 S_2 = Spring load at highest equilibrium speed.

Since the ball arms are parallel to governor axis at the lowest equilibrium speed (i.e. at $N_1 = 290$ r.p.m.), as shown in Fig. 18.20 (a), therefore

$$r = r_1 = 120 \text{ mm} = 0.12 \text{ m}$$

We know that centrifugal force at the minimum speed,

$$F_{C1} = m (\omega_1)^2 r_1 = 2.5 (30.4)^2 \times 0.12 = 277 \text{ N}$$

Now let us find the radius of rotation at the highest equilibrium speed, i.e. at $N_2 = 310$ r.p.m.

The position of ball arm and sleeve arm at the highest equilibrium speed is shown in Fig. 18.20 (b).

Let r_2 = Radius of rotation at $N_2 = 310$ r.p.m.

We know that $h = (r_2 - r_1) \frac{y}{x}$

or
$$r_2 = r_1 + h \left(\frac{x}{y} \right) = 0.12 + 0.015 \left(\frac{0.12}{0.08} \right) = 0.1425 \text{ m}$$

∴ Centrifugal force at the maximum speed,

$$F_{C2} = m (\omega_2)^2 r_2 = 2.5 \times (32.5)^2 \times 0.1425 = 376 \text{ N}$$

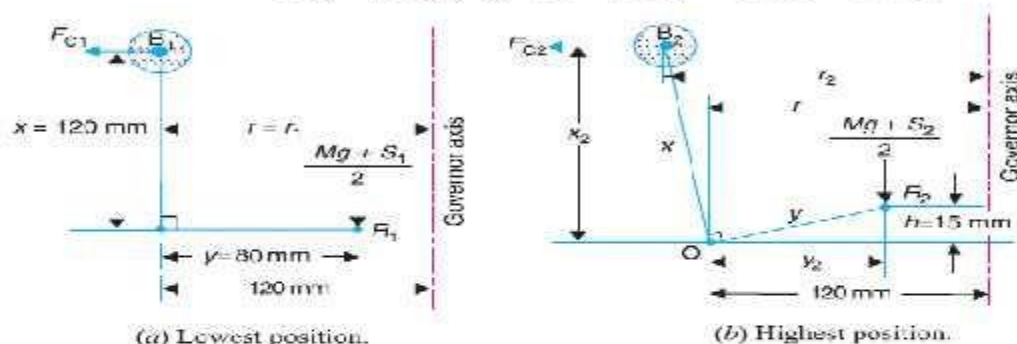


Fig. 18.20

Neglecting the obliquity effect of arms and the moment due to the weight of the balls, we have for lowest position,

$$Mg + S_1 = 2F_{C1} \times \frac{x}{y} = 2 \times 277 \times \frac{0.12}{0.08} = 831 \text{ N}$$

$$\therefore S_1 = 831 \text{ N Ans.} \quad (\because Mg = 0)$$

and for highest position,

$$Mg + S_2 = 2F_{C2} \times \frac{x}{y} = 2 \times 376 \times \frac{0.12}{0.08} = 1128 \text{ N}$$

$$\therefore S_2 = 1128 \text{ N Ans.} \quad (\because Mg = 0)$$

2. **Stiffness of the spring**

We know that stiffness of the spring,

$$s = \frac{S_2 - S_1}{h} = \frac{1128 - 831}{15} = 19.8 \text{ N/mm Ans.}$$

Example 18.14. In a spring loaded Hartnell type governor, the extreme radii of rotation of the balls are 80 mm and 120 mm. The ball arm and the sleeve arm of the bell crank lever are equal in length. The mass of each ball is 2 kg. If the speeds at the two extreme positions are 400 and 420 r.p.m., find : 1. the initial compression of the central spring, and 2. the spring constant.

Solution. Given : $r_1 = 80 \text{ mm} = 0.08 \text{ m}$; $r_2 = 120 \text{ mm} = 0.12 \text{ m}$; $x = y$; $m = 2 \text{ kg}$; $N_1 = 400 \text{ r.p.m.}$ or $\omega = 2 \pi \times 400/60 = 41.9 \text{ rad/s}$; $N_2 = 420 \text{ r.p.m.}$ or $\omega_2 = 2 \pi \times 420/60 = 44 \text{ rad/s}$

Initial compression of the central spring

We know that the centrifugal force at the minimum speed,

$$F_{C1} = m (\omega_1)^2 r_1 = 2 (41.9)^2 \cdot 0.08 = 281 \text{ N}$$

and centrifugal force at the maximum speed,

$$F_{C2} = m (\omega_2)^2 r_2 = 2 (44)^2 \cdot 0.12 = 465 \text{ N}$$

Let S_1 = Spring force at the minimum speed, and

S_2 = Spring force at the maximum speed.

We know that for minimum position,

$$M \cdot g + S_1 - 2 F_{C1} \times \frac{x}{y}$$

$$\therefore S_1 = 2 F_{C1} = 2 \times 281 = 562 \text{ N} \quad \dots (\because M = 0 \text{ and } x = y)$$

Similarly for maximum position,

$$M \cdot g + S_2 = 2 F_{C2} \times \frac{x}{y}$$

$$\therefore S_2 = 2 F_{C2} = 2 \times 465 = 930 \text{ N}$$

We know that lift of the sleeve,

$$h = (r_2 - r_1) \frac{y}{x} = r_2 - r_1 = 120 - 80 = 40 \text{ mm} \quad \dots (\because x = y)$$

$\therefore$ Stiffness of the spring,

$$s = \frac{S_2 - S_1}{h} = \frac{930 - 562}{40} = 9.2 \text{ N/mm}$$

We know that initial compression of the central spring

$$= \frac{S_1}{s} = \frac{562}{9.2} = 61 \text{ mm} \quad \text{Ans.}$$

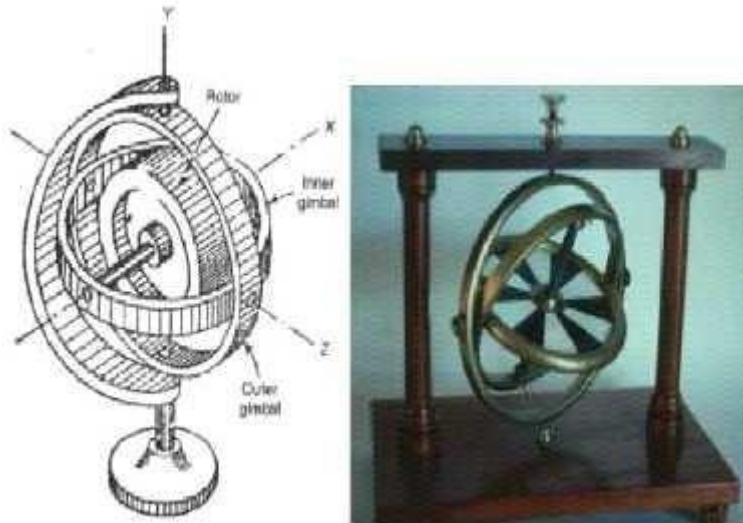
2. Spring constant

We have calculated above that the spring constant or stiffness of the spring,

$$s = 9.2 \text{ N/mm} \quad \text{Ans.}$$

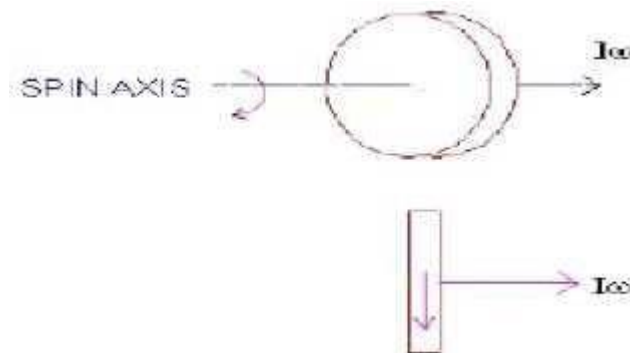
Introduction

'Gyre' is a Greek word, meaning 'circular motion' and Gyration means the whirling motion. A gyroscope is a spatial mechanism which is generally employed for the study of precessional motion of a rotary body. Gyroscope finds applications in gyrocompass, used in aircraft, naval ship, control system of missiles and space shuttle. The gyroscopic effect is also felt on the automotive vehicles while negotiating a turn. A gyroscope consists of a rotor mounted in the inner gimbal. The inner gimbal is mounted in the outer gimbal which itself is mounted on a fixed frame as shown in Fig.. When the rotor spins about X-axis with angular velocity ω rad/s and the inner gimbal precesses (rotates) about Y-axis, the spatial mechanism is forced to turn about Z-axis other than its own axis of rotation, and the gyroscopic effect is thus setup. The resistance to this motion is called gyroscopic effect.



ANGULAR MOTION

A rigid body, (Fig.) spinning at a constant angular velocity ω rad/s about a spin axis through the mass Centre. The angular momentum 'H' of the spinning body is represented by a **vector** whose magnitude is ' $I\omega$ '. I represent the mass amount of inertia of the rotor about the axis of spin.



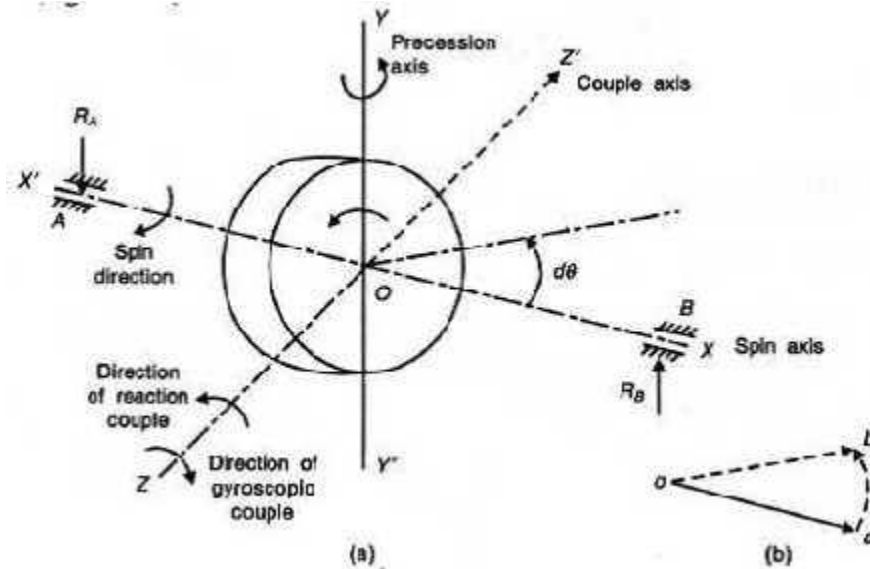
The direction of the angular momentum can be found from the right hand screw rule or the right hand thumb rule. Accordingly, if the fingers of the right hand are bent in the direction of rotation of rotor, then the thumb indicates the direction of momentum.

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GYROSCOPIC COUPLE

Consider a rotary body of mass m having radius of gyration k mounted on the shaft supported at two bearings. Let the rotor spins (rotates) about X-axis with constant angular velocity ω rad/s. The X-axis is, therefore, called spin axis, Y-axis, precession axis and Z-axis, the couple or torque axis (Fig.).

The angular



The angular momentum of the rotating mass is given by,

$$H = mk^2 \omega = I\omega$$

Now, suppose the shaft axis (X-axis) precesses through a small angle $d\theta$ about Y-axis in the plane XOZ, then the angular momentum varies from H to $H + dH$, where dH is the change in the angular momentum, represented by vector ab [Figure 15.2(b)]. For the small value of angle of rotation $d\theta$, we can write However, the rate of change of angular momentum is:

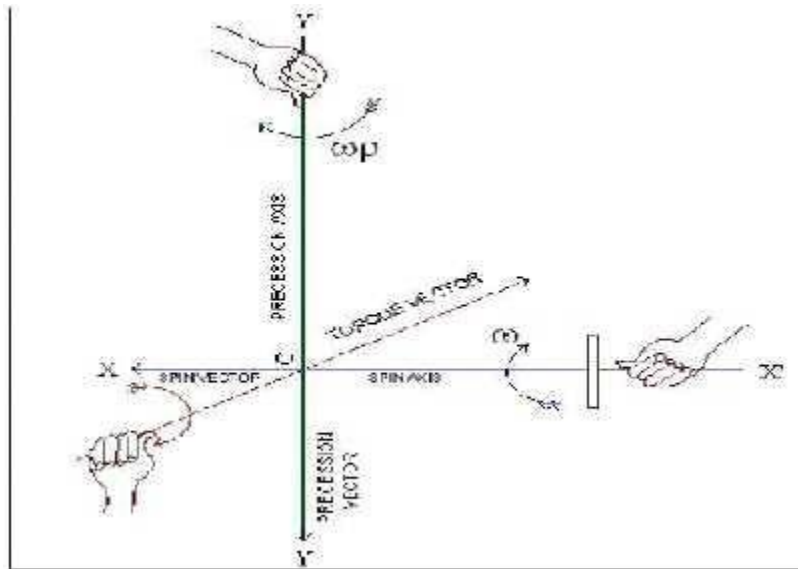
$$C = \frac{dH}{dt} = \lim_{\delta t \rightarrow 0} \left(\frac{I\omega \delta\theta}{\delta t} \right)$$

$$= I\omega \frac{d\theta}{dt}$$

$$C = I\omega\omega_p$$

Direction of Spin vector, Precession vector and Couple/Torque vector with forced precession

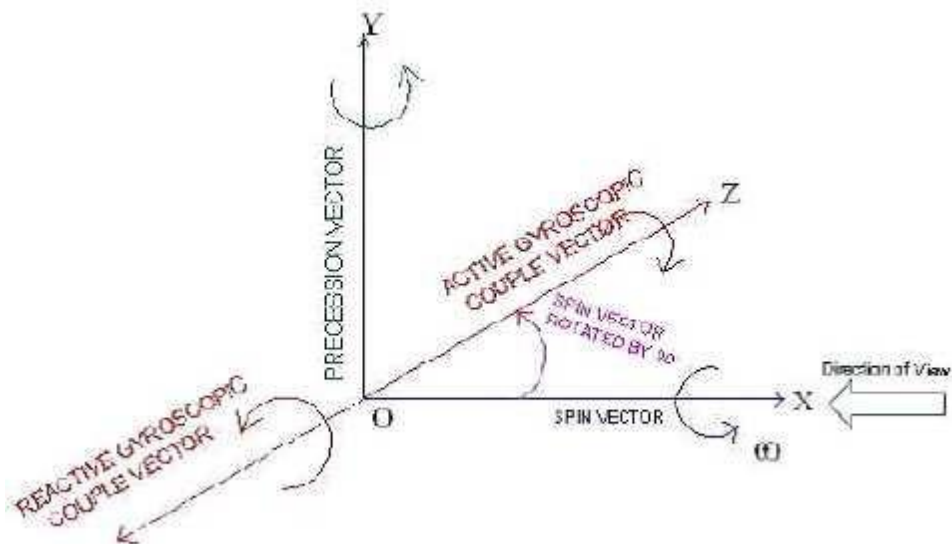
To determine the direction of spin, precession and torque/couple vector, right hand screw rule or right hand rule is used. The fingers represent the rotation of the disc and the thumb shows the direction of the spin, precession and torque vector (Fig.).

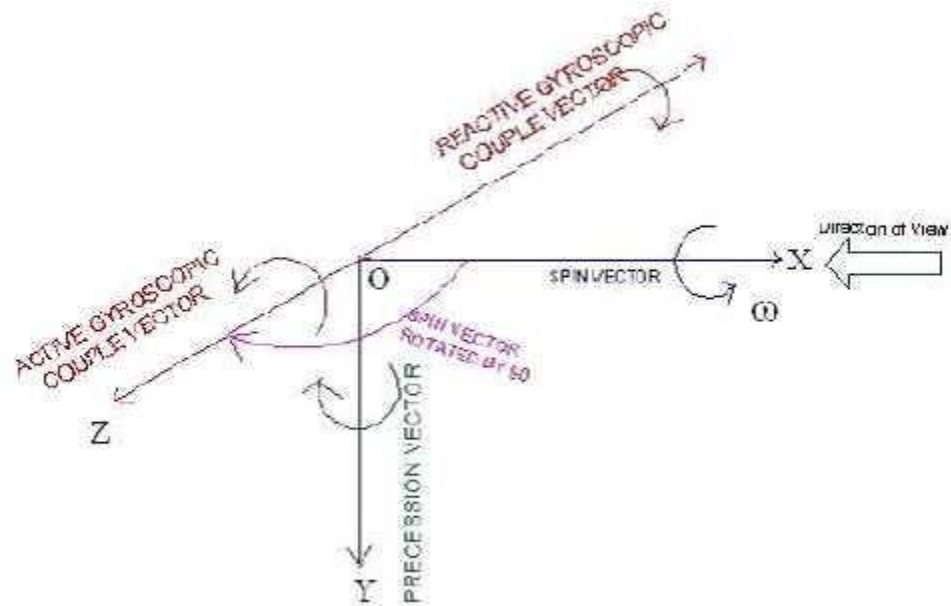


The method of determining the direction of couple/torque vector is as follows

Case (i):

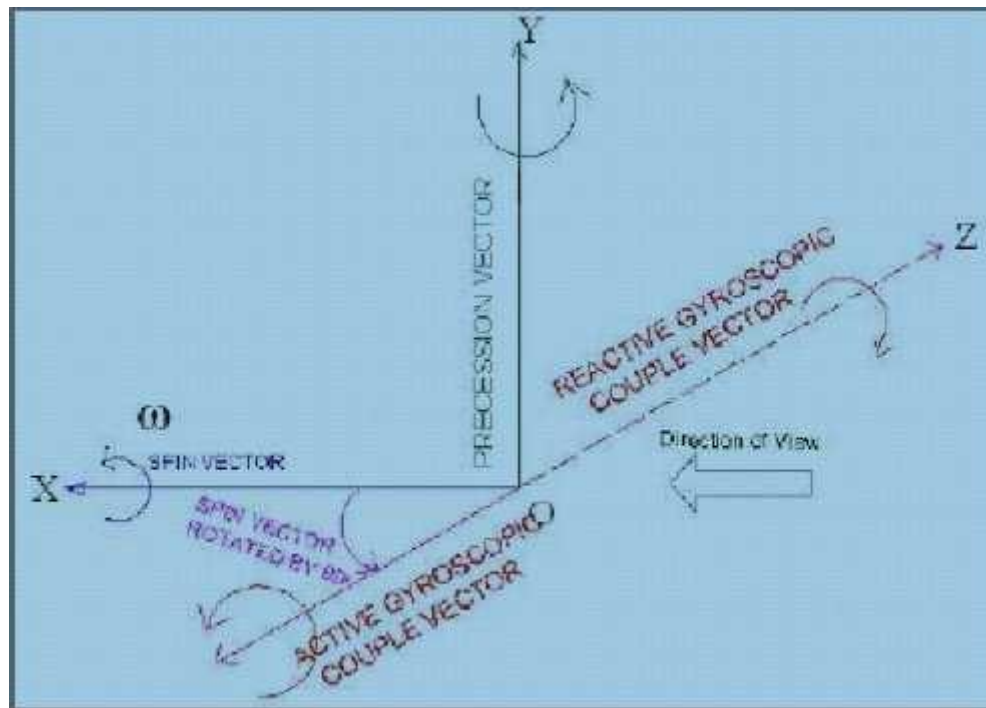
Consider a rotor rotating in anticlockwise direction when seen from the right (Fig.5 and Fig. 6), and to precess the spin axis about precession axis in clockwise and anticlockwise direction when seen from top. Then, to determine the active/reactive gyroscopic couple vector, the following procedure is used. Turn the spin vector through 90° in the direction of precession on the XOZ plane. The turned spin vector will then correspond to the direction of active gyroscopic couple/torque vector. The reactive gyroscopic couple/torque vector is taken opposite to active gyro vector direction

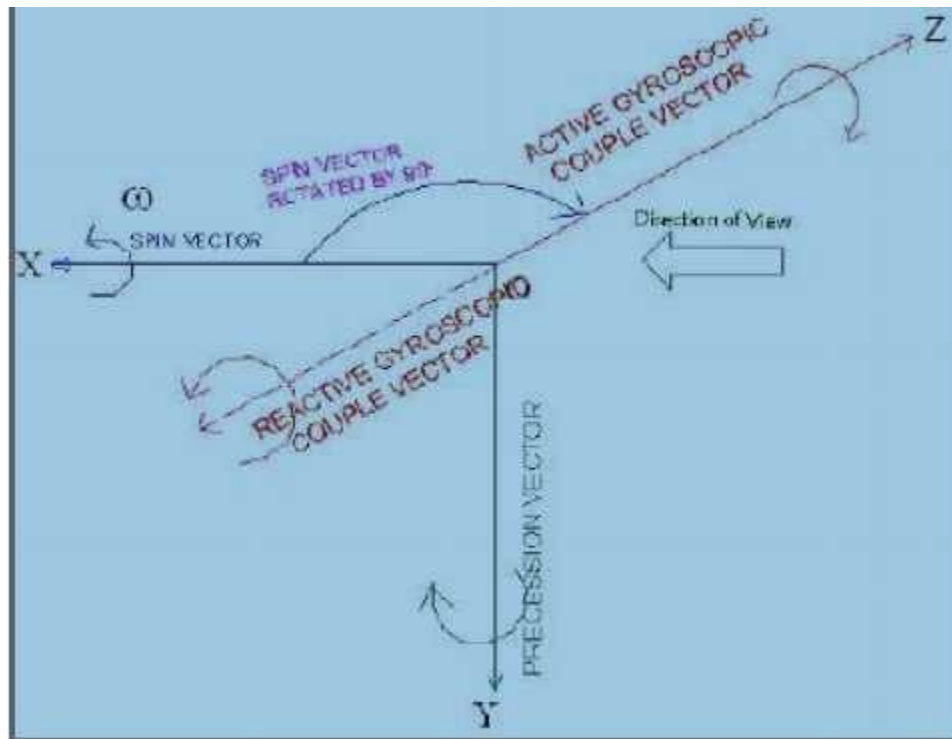




Case (ii):

Consider a rotor rotating in clockwise direction when seen from the right (Fig.7 and Fig. 8), and to precess the spin axis about precession axis in clockwise and anticlockwise direction when seen from top. Then, to determine the active/reactive gyroscopic couple vector, turn the spin vector through 90° in the direction of precession on the XOZ plane. The turned spin vector will then correspond to the direction of active gyroscopic couple/torque vector. The reactive gyroscopic couple/torque vector is taken opposite to active gyro vector direction





The resisting couple/ reactive couple will act in the direction opposite to that of the gyroscopic couple. This means that, whenever the axis of spin changes its direction, a **gyroscopic couple** is applied to it through the bearing which supports the spinning axis.

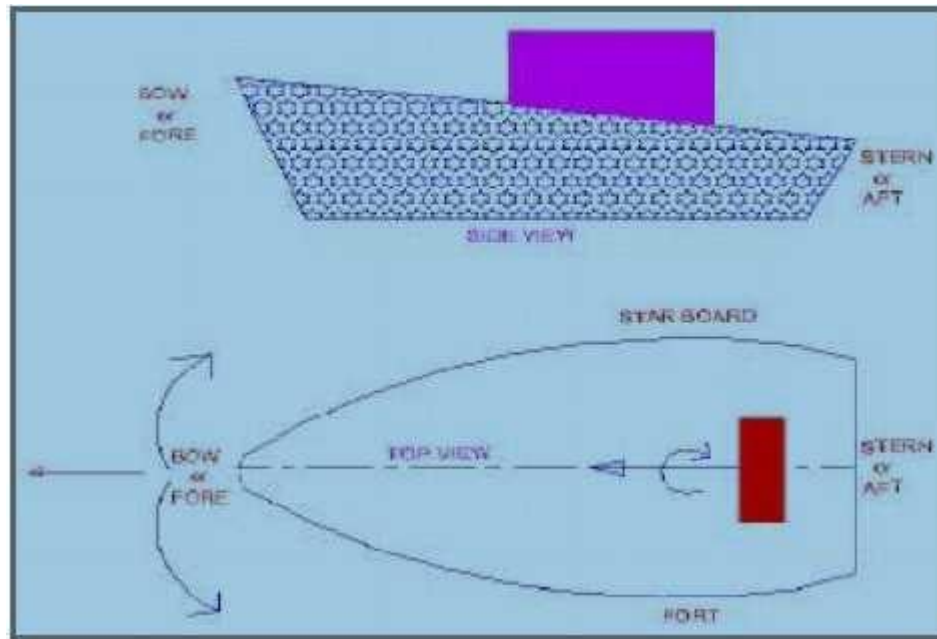
GYROSCOPIC EFFECT ON SHIP

Gyroscope is used for stabilization and directional control of a ship sailing in the rough sea. A ship, while navigating in the rough sea, may experience the following three different types of motion:

- (i) Steering—the turning of ship in a curve while moving forward
 - (ii) Pitching—the movement of the ship up and down from horizontal position in a vertical plane about transverse axis
 - (iii) Rolling—Sideway motion of the ship about longitudinal axis
- For stabilization of a ship against any of the above motion, the major requirement is that the gyroscope shall be made to precess in such a way that reaction couple exerted by the rotor opposes the disturbing couple which may act on the frame.

Ship Terminology

- (i) Bow – It is the fore end of ship
- (ii) Stern – It is the rear end of ship
- (iii) Starboard – It is the right hand side of the ship looking in the direction of motion
- (iv) Port – It is the left hand side of the ship looking in the direction of motion

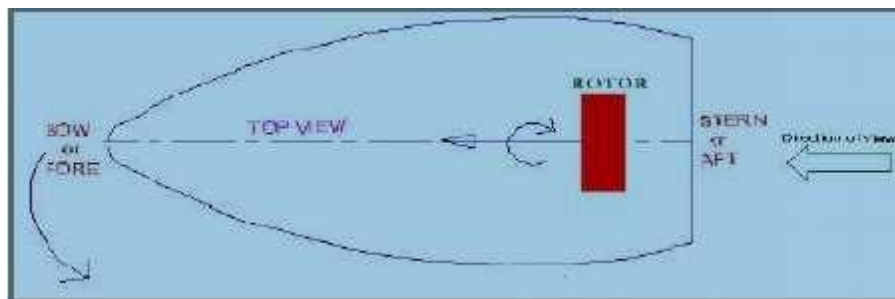


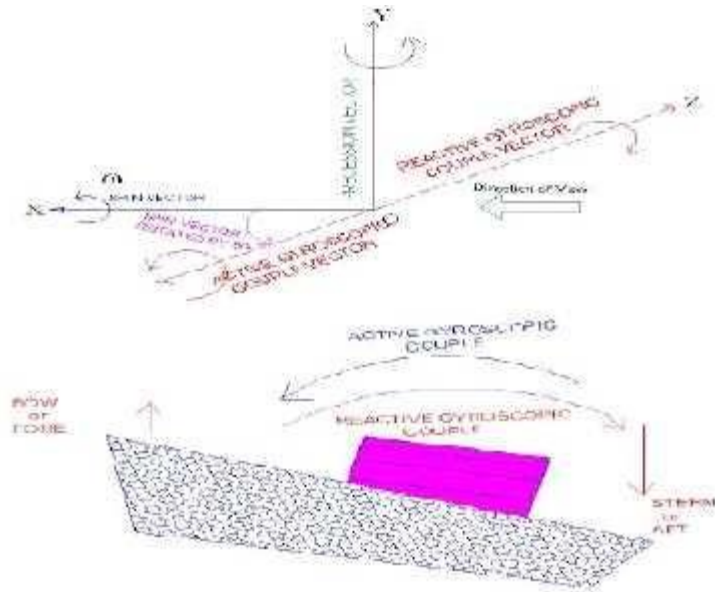
Consider a gyro-rotor mounted on the ship along longitudinal axis (X-axis) as shown in Fig.10 and rotate in clockwise direction when viewed from rear end of the ship. The angular speed of the rotor is ω rad/s. The direction of angular momentum vector oa , based on direction of rotation of rotor, is decided using right hand thumb rule as discussed earlier. The gyroscopic effect during the three types of motion of ship is discussed.

Gyroscopic effect on Steering of ship

(i) Left turn with clockwise rotor

When ship takes a left turn and the **rotor rotates in clockwise direction** viewed from stern, the gyroscopic couple act on the ship is analyzed in the following way.





Note that, always reactive gyroscopic couple is considered for analysis. From the above analysis (Fig.), the couple acts over the ship between stern and bow. This reaction couple tends to raise the front end (bow) and lower the rear end (stern) of the ship.

(ii) Right turn with clockwise rotor

When ship takes a right turn and the **rotor rotates in clockwise direction** viewed from stern, the gyroscopic couple acts on the ship is analyzed (Fig 14). Again, the couple acts in vertical plane, means between stern and bow. Now the reaction couple tends to lower the bow of the ship and raise the stern.